

CITYZEN PROJECT TBILISI, GEORGIA

LEED Consultant's REPORT

**for LEED Certification Preliminary Assessment for
Citizen Development**

Stage: Feasibility

Produced by: Sustainable Engineering Ltd

Content:

Key Targets and Strategies

Owner's Capacity and Intention

General Action Plan

Further analysis, Assessments, Consultancy or Other Service Which May Be
Needed:

General Assessment by Sustainability Categories

Estimated Credit Options

Annex A - Pathway

Updated: November 2024 V 2.0

Introduction

The site is located in central Tbilisi, Georgia, on previously developed land (a former military site). The gross project floor area (GFA) is approximately 60,000 m². The initial conceptual design was created by Zaha Hadid Architects, with local design contributions from Artstudio Project. Final design and engineering documentation will be developed by various engineering firms. The client aims to achieve a **Silver** or **Gold** level LEED certification for the site. The consultant recommends applying the following requirements to the entire project.

B Owners Project Requirements

General Part

This section includes the finalized Owner's Project Requirements (OPR) for the facility, initially developed during the Pre-Design Phase and subsequently updated by the Owner, LEED Consultant, and, later, the Commissioning Authority during the LEED certification process..

2. Owner and User Requirements

A. **Cityzen Project** will develop a mixed use building. It will be conceived as space for residential, office and other purposes. The building shall be designed as an environmentally friendly "green" structure. The building shall include innovative green features for energy efficiency, water efficiency, indoor environmental quality, waste reduction among others that will benefit the environment in all stages of construction and operation.

B. The functionality of the building should not be compromised while making the building green. The building orientation and layout should be optimized per the general requirements of the Client as identified in the specifications.

3. Environmental and Sustainability Goals

- The project will meet **LEED v4** or **v4.1** requirements for building design and construction, aiming for Silver or Gold performance levels.
- Priorities include site sustainability, water efficiency, energy efficiency, and indoor air quality.
- Modern architectural design, along with advanced HVAC and electrical systems, will ensure energy efficiency.
- Renewable energy solutions such as solar collectors, solar photovoltaics, and solar tubes will provide a significant portion of the building's energy needs.
- Water efficiency strategies will reduce water consumption below the baseline required by LEED standards.

4. Energy Efficiency Goals

A. Energy Efficiency Targets:

B. Exceed **ASHRAE 90.1-2010** energy efficiency standards by 20–32%, verified through energy simulation.

C. Lighting Systems:

D. Implement cost-effective, energy-efficient lighting fixtures (preferably LED) with daylight control features.

E. HVAC Systems:

F. Utilize high-efficiency HVAC systems exceeding ASHRAE standards by at least 25%.

G. Building Envelope:

H. Optimize insulation, window selection, and piping/ductwork insulation to reduce energy demand.

I. Building Management System (BMS):

J. Install a comprehensive, user-friendly BMS for HVAC and lighting controls..

5. Indoor Environmental Quality Requirements

- Comply with **EN 12464** lighting standards.
- Provide individual controls for multi-occupant spaces (occupancy sensors and manual overrides).
- Ensure thermal comfort per **ASHRAE 55-2004**.
- Exceed ventilation standards (**ASHRAE 62.1-2010**) by at least 30%.
- Use air filters with a minimum MERV 13 (F7) efficiency rating.
- Equip individual offices and work areas with independent thermostats.
- Install lighting systems with multi-mode functionality (e.g., general and A/V presentation modes).
- Maintain acoustical comfort as per industry standards.

6. Equipment and Systems Expectations

- State-of-the-art HVAC systems shall be used where feasible. These systems may include but not limited, high efficient chiller and boilers, thermal storage systems, passive systems, heat recovery systems etc.,
- Renewable energy technologies shall be applied, e.g. solar collector – water heaters, PV. Also passive systems such as Natural Ventilation, Earth Labyrinth systems may be included in order to decrease fossil based energy consumption.
- The use of low temperature heating and cooling such as chilled beam, and TABS can be preferred.
- State-of-the-art lighting systems shall be used throughout the building. These include but not limited to tandem electronic ballast T5 or LED lighting fixtures, occupancy/daylight sensors, zoning, timing etc.
- Inefficient systems and equipment shall not be accepted.
- Dimmable lighting fixtures shall be used only where feasible.
- Generators shall supply 100 % energy for office areas lighting and power sockets.
- Cost optimization should be made during selection and design of the system.

7. Building Occupant and O&M Personnel Expectations

- Day-to-day HVAC operation shall be done by Citizen Project Operation and Maintenance (O&M) staff. Proper demonstration training shall be provided to O&M staff after the installation of systems.
- Equipment manufacturers shall include periodic maintenance in their proposal for the warranty period.
- The systems shall be properly commissioned and a commissioning report along with a system manual shall be prepared.
- Selected systems should be innovative yet easy to use and explain to building management.

8. Estimated Schedule of Construction

Design phase shall be completed by March-April 2025. Construction shall be started until May 2025. Building shall be completed until May 2026.

9. Other

A. The specification which were given in annex J of ASHRAE Guideline 1.1-2010, "*HVAC&R Technical Requirements for The Commissioning Process*" is also part of this document.

Codes, Regulations And Sources

Some examples of the documents which will be used in this Project are given below.

- [1] ASHRAE Standard 189.1, Standard for the Design of High-Performance Green Buildings.
- [2] ASHRAE Standard 90.1, Energy Standard for Buildings Except-Low Rise Residential Buildings and Appendix G. [3] ASHRAE Standard 62.1, Ventilation for Acceptable Indoor Air Quality.
- [4] ASHRAE Standard 55-2004, Thermal Environmental Conditions for Human Occupancy. [5] ASHRAE Guideline 0-2005, "The Commissioning Process".
- [6] ASHRAE Guideline 1.1-2007, "HVAC&R Technical Requirements for The Commissioning Process".
- [7] LEED Guidebooks.

C Strategies to achieve key targets

LOCATION AND TRANSPORTATION

- Provide sufficient compliant facilities for bicycle users
- Provide access to mass transit and surrounding facilities
- Develop the site in densely developed location
- Minimize vehicle journeys to the site, reduce number of parking places.
- Stimulate use of alternative environmentally friendlier transportation

SUSTAINABLE SITES

- Minimize environmental pollution during the construction phase
- Provide the most environmentally friendly site design solutions based on preliminary environmental assessment, protect the existing habitat
- Reduce the building footprint and maximize the open space.
- Minimize rain water run-off from the site,

- Protect the habitat by reducing overheating of the hardscape
- Provide lighting minimizing light trespass out of the site and to the sky

WATER EFFICIENCY

- Minimize use of potable water for irrigation
- Minimize use of water by indoor items, apply low water use rate fixtures and technologies
- Provide measurement of consumed water by sub metering of the different key users
- Avoid excess use of water by cooling towers

ENERGY AND ATMOSPHERE

- Conduct commissioning of building fabric and all its systems, verify performance and compare with initial targets.
- Minimize area of the building envelope, apply high decrement factor and thermal mass materials, use low thermal conductivity materials.
- Maximize solar gains in winter and minimize it in summer.
- Maximize use of daylight during occupancy hours.
- Use efficient lighting appliances
- Use environmentally friendly refrigerants
- Provide measurement of consumed energy by sub metering of the different key users
- Produce maximum renewable energy on the site
- Provide maximum recovery of exhaust air (ventilation)
- Use efficient heat generation systems like boiler, heat pump
- Apply BMS and other thermal/air management systems

MATERIALS AND RESOURCES

- Maximize reuse of building materials by retaining the existing structure of the building/s, use of the materials from the previous site development or supply of reused materials, divert construction waste for re-use or recycling.
- Maximize use of recycled materials and products
- Use lower life cycle impact materials, rapidly renewable materials and locally produced materials, ecologically labeled materials

INDOOR ENVIRONMENTAL QUALITY

- Increase indoor air quality by using low emissive materials, also applying post construction flush out and measurement strategies
- Prevent distribution of tobacco smoke in the building by designing designated compliant areas
- Provide thermal, lighting and acoustical comfort
- Provide sufficient quality fresh air for occupants
- Provide views to outside and proper daylight to most of the spaces and occupied areas.

Owner's Capacity and Intention

The owner of the project is Cityzen Development having its project management office in Tbilisi, Georgia. The developer applies both – international and local expertise in architecture and engineering. The developer co-operates with teams of overseas and local architects, as well as teams of local engineers. Owner intends to develop certified building for two reasons – corporate sustainability policy and marketability of the developed property.

General Action Plan

Activity/ months	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6+
Finalization of architectural concept	x	x	x										
Formation of OPR	x	x											
Formation of BoD		x	x	x									
Identification of initial building services strategies	x	x											
Energy modeling				x	x			x	x	x			
Adjustment of the architectural design and building services according to energy model and credits requirements						x	x	x	x	x	x		x
Planning for construction processes			x	x	x	x	x	x	x	x	x		
Review and if necessary modification of tender documents													x
Supervision of construction process													x
Commissioning												x	x
General meetings and communication with designer, building services and LEED teams	x	x	x	x	x	x	x	x	x	x	x	x	x

Further analysis, assessments, consultancy or other service which may be needed:

Description	Content	Comment
Commissioning service	Commissioning of basic building systems	Can be owners team member, ideally the future facilities manager or engineer.
Enhanced Commissioning service	Commissioning of other systems and beyond the occupancy	Must be an independent entity. In this case can be signed off by LEED Expert, but implemented by entity.
HVAC Engineering	Design of the building systems	According to ASHRAE requirements and energy simulation recommendations
Lighting Engineering	Design of the interior and exterior lighting	According to light pollution prevention and indoor environmental quality requirements
Water supply, irrigation and drainage engineering	Design of indoor water use and irrigation systems, storm water management system	Efficient water use requirements, on-site storm water management facilities
Landscape survey and habitat restoration works	Restoration of the greenfield portions of the site	Some of the works, like topography, is already done
Acoustic service	Acoustical properties of spaces, reverberation times, sound masking and amplification strategies.	Can be outsourced to acoustical studio.
Air Testing	After construction and before occupancy internal air quality testing	As by ASTM standard, EPA compendium method, or ISO method By accredited laboratory as by ISO/IEC 17025, for chemical analysis tests. Perform by Accredited professional
Site Assessment	Additional site survey is needed	Can be outsourced, in addition to existing site survey works done
Brownfield Assessment	ESA 1, ESA 2, remediation works	Needed also request and acceptance from the authorities.
Life Cycle Impact assessment	After materials list is produced	Can be outsourced
Other, as needed according to client's needs		

General Assessment by Sustainability Categories:

Category	Comment
Integrative Process	Project owner has mobilized international and local teams of architects, building services engineers, cost estimators, sustainability advisers.
Sustainable Sites	The project is located in the averagely developed area, because of surrounding parks, cemetery. However, the area is very well accessible by public transport, and cyclists network. The project has potential to develop maximum environmentally friendly site design and adhere light pollution, storm water management and other issues. The site was previously occupied by military institution, so, it can be contaminated and classified as brownfield. The owner is ready to conduct a

	study and approach local authorities to accept their effort for brownfield remediation. The site has capacity to maximize open space, restore natural habitat. Taking into account recent development of electric car charging network in Tbilisi, the project plans to equip the new development with charging facilities.
Water Efficiency	Project has capacity to apply low water use fixtures at all locations. Additional water gains is planned to get through foundation drain which can be counted towards reclaimed water, as it would anyway flow and be pumped out. Project can apply low watering need species of trees and shrubs.
Energy and Atmosphere	Project is applying different strategies for energy use reduction such as efficient heat generators, efficient heat pumps, ventilation air heat recovery, building management system. It is planned to minimize transparent part of the façade and minimize thermal conductivity of the envelope applying high decrement factor materials. Modern heat pumps will minimize refrigerant charge and save energy. Project plans to apply commissioning agent's service.
Materials and Resources	This is the hardest category for the project as local market does not offer yet certified materials. Vast majority of possible green certified (different types of certificates) are to be imported and are rather expensive, which creates risk to achieve materials part credits. However, in terms of waste management there is good practice in the country.
Indoor Environmental Quality	Project design considers good views out, good daylight factors for permanently occupied spaces. Thermal comfort is to be achieved through good MEP design. Local HVAC and lighting engineers have capacity to design as by ASHRAE requirements and good lighting practice. In acoustics part, taking into account previous practice in Georgia, the problem may be achievement of required reverberation time in large spaces (in non-residential sections).

Estimated Credit Options

Prerequisites: YES

Credits: „YES“ – 51 „MAY BE“ - 27 „No“ - 31

Total minimum – 50 corresponds to level “SILVER”

Total estimated (YES + 30% of “may Be”) - 60 - “GOLD”

Possible Rating Systems: LEED ND+C or LEED Homes (multi-story apartments option)

Annex A – Pathway (Review of Credit options)

CITIZEN SITE TBILISI, GEORGIA

Y	"?"	N					Responsible Engineer	Action Required	Possible cost	Applicability, Feasibility, other comment	
1	0	0	Credit 1	Integrative Process			1	Project Manager, all engineers	Synergy of all key management and engineering staff, group work as needed	No cost increase predicted	Feasible
			Location and Transportation			Possible Points:	16				
		x	Credit 1	LEED for Neighborhood Development Location			16	n/a	n/a	n/a	Not aiming to take the credit
1	0	0	Credit 2	Sensitive Land Protection			1	Project Manager, Architect	Documenting of land properties	No major associated cost	Planned credit
0	2	0	Credit 3	High Priority Site			2	Project Manager, Architect	n/a	Very high cost for Brownfield option	Upon Client's Decision
3	0	2	Credit 4	Density and Diverse Uses Surrounding			5	Project Manager, Architect	Documenting the existing surrounding	No major associated cost	Planned credit
4	1	0	Credit 5	Access to Quality Transit			5	Project Manager, Architect	Documenting the routs and schedules	No major associated cost	Planned credit
1	0	0	Credit 6	Bicycle Facilities			1	Architect	n/a	n/a	No bicycle network
1	0	0	Credit 7	Reduced Parking Footprint			1	Architect	Collection of documentation, parking design	No major associated cost	Planned credit
1	0	0	Credit 8	Green Vehicles			1	Architect, Other engineers, if special equipment is involved	Showing plans, assigning parking	No extra cost seen	Parking can be reduced
11	3	2					16				
			Sustainable Sites			Possible Points:	10				

Y			Prereq 1	Construction Activity Pollution Prevention	Required	Project Manager, Architect, construction Management engineer (In this case with support of LEED Consultant)	Plan and documenting on construction stage, contractors' training	Minor increase in construction cost	Required
1	0	0	Credit 1	Site Assessment	1	Project Manager, Architect, construction Management engineer (In this case with support of LEED Consultant)	Collection of existing data, adding new data	Minor increase in construction cost	Planned credit
1	0	1	Credit 2	Site Development--Protect or Restore Habitat	2	Architect, Landscape Engineer	To achieve 30% native green (incl. roofs)	Increase in designed cost but within feasible margins	Planned credit
1	0	0	Credit 3	Open Space	1	Architect, Landscape Engineer	Provide 30% open space and 25% vegetated space	Increase in designed cost but within feasible margins	Planned credit
0	1	2	Credit 4	Rainwater Management	3	Architect, Landscape Engineer, Water supply engineer	Obtaining local data, comparing with the Act values	Increase in designed cost but within feasible margins	Planned credit
2	0	0	Credit 5	Heat Island Reduction	2	Architect, Materials Engineer	Selection of compliant materials, covers and green roof	Increase in designed cost but within feasible margins	Planned credit
1	0	0	Credit 6	Light Pollution Reduction	1	Architect, Power Supply Engineer	Compliant lighting design	Increase of project cost but within feasible margins	Planned credit
6	1	3			10				
			Water Efficiency		Possible Points: 11				
Y			Prereq 1	Outdoor Water Use Reduction	Required	Architect, Landscape Engineer, Water Supply Engineer	Provide no watering or low watering plants, and effective watering items.	Feasible increase of the project cost	Required
Y			Prereq 2	Indoor Water Use Reduction	Required	Water Supply Engineer	Design of baseline compliant or better water supply items and water consuming appliances	Minor increase of the project cost	Required
Y			Prereq 3	Building-Level Water Metering	Required	Water Supply Engineer	Design whole site water meter	Neglegable increase of project cost	Required
1	1	0	Credit 1	Outdoor Water Use Reduction	2	Architect, Landscape Engineer, Water Supply Engineer	Provide no watering or low watering plants, and effective watering items.	Feasible increase of the project cost	Planned credit

5	1	0	Credit 2	Indoor Water Use Reduction	6	Water Supply Engineer	Design of baseline compliant or better water supply items and water consuming appliances	Minor increase of the project cost	Planned credit
0	0	2	Credit 3	Cooling Tower Water Use	2	HVAC Engineer	n/a	n/a	Credit can not be achieved
1	0	0	Credit 4	Water Metering	1	Water Supply Engineer	Design compliant water sub meters	Negligible increase of project cost	Planned credit
7	2	2				11			
			Energy and Atmosphere		Possible Points:	33			
Y			Prereq 1	Fundamental Commissioning and Verification	Required	Project Manager, Architect, commissioning agent	Form the OPR and BOD, hire commissioning agent	Do not increase construction cost, increases management cost	Required
Y			Prereq 2	Minimum Energy Performance	Required	Project Manager, Architect, HVAC Engineer, Materials Engineer, Power supply Engineer, Environmental Engineer	Finalize space design, apply suggested envelope properties and shading devices, renewable energy technologies. Design the HVAC system, lighting. Initial check for compliance	Increase of project total cost, feasible within the whole building life cycle	Required
Y			Prereq 3	Building-Level Energy Metering	Required	Power Supply Engineer	Design building level energy meter	Negligible increase of project cost	Required
Y			Prereq 4	Fundamental Refrigerant Management	Required	HVAC Engineer	Design with no CFC refrigerant	No increase of the project cost	Required
	2	4	Credit 1	Enhanced Commissioning	6	Project Manager, Architect, commissioning agent	Recruiting commissioning authority	Some increase in management cost	Upon Client's decision
8	2	8	Credit 2	Optimize Energy Performance	18	Project Manager, Architect, HVAC Engineer, Materials Engineer, Power supply Engineer, Environmental Engineer	Finalize space design, apply suggested envelope properties and shading devices, renewable energy technologies. Design the HVAC system, lighting. Initial check for compliance	Increase of project total cost, feasible within the whole building life cycle	Planned credit
1	0	0	Credit 3	Advanced Energy Metering	1	Power Supply Engineer	Design compliant power meters and data storage	Minor increase of the project cost	Planned credit

1	1	0	Credit 4	Demand Response	2	Project Manager, Power supply Manager, HAC Engineer, Water supply Engineer (N/a in Georgia)	N/A	n/a	Not aiming to take the credit
0	1	2	Credit 5	Renewable Energy Production	3	Project Manager, Architect, HVAC Engineer, Materials Engineer, Power supply Engineer	Design renewable energy systems	Sufficient but justified increase of the project cost	Planned credit
1	0	0	Credit 6	Enhanced Refrigerant Management	1	Project Manager, Architect, HVAC Engineer	ODP and GWP data from designer, calculations of values	Minor increase of the project cost	Planned credit
0	0	2	Credit 7	Green Power and Carbon Offsets	2	Project Manager, Electrical engineer (n/a in Georgia)	n/a	n/a	Not aiming to take the credit
11	6	16	33						
			Materials and Resources	Possible Points:	13				
Y			Prereq 1	Storage and Collection of Recyclables	Required	Architect, Processes Engineer (technologist)	Consider adequate space in design and furniture design	Negligible increase of project cost	Required
Y			Prereq 2	Construction and Demolition Waste Management Planning	Required	Project Manager, Construction Process Engineer, construction contractor/s	Consider in construction management plan, finding recycling contractors	Negligible increase of project cost	Required
0	2	3	Credit 1	Building Life-Cycle Impact Reduction	5	Project Manager, Architect, Environmental engineer	Reused materials are to be designed and used	Only partitions and exterior walls (partially) can be counted	Feasibility is to be confirmed
0	1	1	Credit 2	Building Product Disclosure and Optimization - Environmental Product Declarations	2	Project Manager, Architect, Materials Engineer, Procurement Officer	Design materials schedule accordingly	Sufficient increase of the project cost	Feasibility is to be confirmed
0	1	1	Credit 3	Building Product Disclosure and Optimization - Sourcing of Raw Materials	2	Project Manager, Architect, Materials Engineer, Procurement Officer	Design materials schedule accordingly	Sufficient increase of the project cost	Feasibility is to be confirmed
0	1	1	Credit 4	Building Product Disclosure and Optimization - Material Ingredients	2	Project Manager, Architect, Materials Engineer, Procurement Officer	Design materials schedule accordingly	Sufficient increase of the project cost	Feasibility is to be confirmed
1	1	0	Credit 5	Construction and Demolition Waste Management	2	Project Manager, Construction Process Engineer, Procurement Officer	n/a	The building is demolished	Not aiming to take the credit
1	6	6	13						

			Indoor Environmental Quality	Possible Points:	16			
Y			Prereq 1 Minimum Indoor Air Quality Performance	Required	Architect, HVAC Engineer	To determine minimum flow and fresh air rates, design according to this	Mandatory	Required
Y			Prereq 2 Environmental Tobacco Smoke Control	Required	Architect	Design of signs	Mandatory	Required
2	0	0	Credit 1 Enhanced Indoor Air Quality Strategies	2	Architect, HVAC Engineer	Design entryway systems and HVAC according to, design negative pressure at contaminated rooms and cross contamination from garage and toilets, design filtration or prohibit smoking	Feasible	Planned Credit
1	1	1	Credit 2 Low-Emitting Materials	3	Architect, Materials Engineer	Formation of requirements for each material	Depends on availability of materials	Planned partially
1	0	0	Credit 3 Construction Indoor Air Quality Management Plan	1	Construction Process Engineer, Construction contractor/s	To incorporate requirements in the construction management plan, to make contractor follow the plan and get verification	Minor increase of the cost	Planned credit
1	1	0	Credit 4 Indoor Air Quality Assessment	2	Project Manager, Environmental Engineer, construction contractor	To plan for flush out, implement before occupancy	To check with client	Planned credit
0	1		Credit 5 Thermal Comfort	1	Architect, HVAC Engineer	very hard for open	Cost increase	not realistic
2	0	0	Credit 6 Interior Lighting	2	Architect, Power supply Engineer	Conduct relevant power lighting design	Feasible, increase of the cost	Planned credit
1	1	1	Credit 7 Daylight	3	Architect, environmental Engineer (may be LEED Consultant)	finalizing design, measuring daylight	Feasible	Feasible credit
0	1	0	Credit 8 Quality Views	1	Architect	finalizing design, measuring views	Feasible	Feasible credit
0	1	0	Credit 9 Acoustic Performance	1	Architect, Environmental Engineer (may be LEED Consultant)	Designing absorptive materials	Feasible	Feasible credit
8	6	2	16					
			Innovation	Possible Points:	6			

2	2	0	Credit 1	Innovation	5	Project Manager, Architect, other staff case by case			
1	0	0	Credit 2	LEED Accredited Professional	1	In-house team member or LEED Consultant			Planned credit
3	2								
			Regional Priority		Possible Points:	4			
1	0	0	Credit 1	Regional Priority: Specific Credit	1	Project Manager, Architect, other staff case by case	Energy performance		Planned credit
1	0	0	Credit 2	Regional Priority: Specific Credit	1	Project Manager, Architect, other staff case by case	Sensitive land		Planned credit
1	0	0	Credit 3	Regional Priority: Specific Credit	1	Project Manager, Architect, other staff case by case	Light pollution reduction		Planned credit
0	1	0	Credit 4	Regional Priority: Specific Credit	1	Project Manager, Architect, other staff case by case	Restore habitat		
3	1	0					Rainwater management		
51	27	31	Total		Possible Points:	110			

Certified 40 to 49 points Silver 50 to 59 points Gold 60 to 79 points
 Platinum 80 to 110